

MATH 1210 Worksheet 3 Summer 2023

Suggested exercises, more exercises can be found in the textbook or on Dr. R.S.D. Thomas' webpage for archived material:

<http://home.cc.umanitoba.ca/~thomas/Courses>

1. Write the following in Cartesian form.

(a) $e^{-i\pi/2}$

(b) $4 \frac{e^{2i\pi/3}}{e^{i\pi/6}}$

2. Write

$$\overline{(2 + \sqrt{2}e^{i5\pi/4})} \cdot \frac{1}{1 - e^{i\pi/3}}$$

in exponential form. Use the principal value for the argument.

3. Find all the complex numbers z solving the equation

$$z^5 = -16\sqrt{2} - 16\sqrt{2}i.$$

Write them in exponential form and use principal value of their arguments.

4. Find all the solution of the equation $x^6 - 4x^3 + 8 = 0$. Write the roots in exponential form and use the principal value of their arguments.
5. Use polynomial division to divide $P(x) = 7x^5 - 11x^4 - 8x^3 + 9x^2 - 13x + 6$ by $x - 2$. Deduce that 2 is a root.
6. Use polynomial division to divide $P(x) = x^4 + x^3 + 3x^2 - x + 1$ by $x + 1$. Deduce that -1 is not a root.
7. If $-2 + 3i$ is a zero of the polynomial $P(x) = x^3 + 7x^2 + 25x + 39$, find its other zeros.
8. For what value(s) of k will the polynomial $P(x) = 5x^5 + 4x^4 + kx^3 + 2x^2 + x + 1$ have remainder 15 when divided by $x - 1$.
9. Find the remainder when $P(x) = x^{999} + 2x^{998} + x^2 - 1$ is divided by $x + 1$.
10. Consider the polynomial

$$P(x) = 4x^5 + 3x^4 - 2x^3 + x^2 - x - 2.$$

- (a) Use Descartes' rule of signs to state the number of possible positive and negative solutions.
- (b) Use Bounds theorem find a bound on the roots.
- (c) Use the Rational Root Theorem to list all possible rational solutions. Use the results from the first two parts.
11. Consider the polynomial

$$P(x) = x^3 + 4x^2 + 12x + 9.$$

- (a) Show that it has no zeroes in the interval $[-100, -20]$.
- (b) Show that it has no zeroes in the interval $[0, 13]$.
12. Consider the polynomial

$$P(x) = 2x^4 + 2x^3 + x^2 + 5x - 10.$$

- (a) Use the Rational Root Theorem to determine all possible rational roots.

- (b) Let $Q(x) = (x - 5)P(x)$. How many positive real roots and negative real roots does $Q(x)$ have? Explain.
- (c) Use Bounds Theorem to determine an upper bound for the modulus of roots of $P(x)$. Does this eliminate any possible rational roots? Which ones?

13. Consider the polynomial

$$g(x) = x^3 - mx^2 - 1$$

where m is an integer. Find all values of m for which $g(x)$ has a rational root.